Algorithm for file updates in Python

Project description

At the organization, a file named allow_list.txt identifies IP addresses that are allowed within the network. Another remove_list contains IP addresses that need to be removed from the allow_list so an algorithm was created to iterate through these lists and remove the IP addresses contained in the remove_list from the allow_list

Open the file that contains the allow list

```
import_file = "allow_list.txt"
remove_list = {}
with open(import file, "r") as file:
```

For the first part of the algorithm the allow_list is imported into the system, the remove list is created (empty as a placeholder), and we open the allow_list as a read file.

Read the file contents

```
ip_addresses = file.read().split()
```

As the next step, we read the contents of the file.

Convert the string into a list

The split() command in the above screenshot splits the read contents of the file into a list.

Iterate through the remove list

```
for element in remove_list:
```

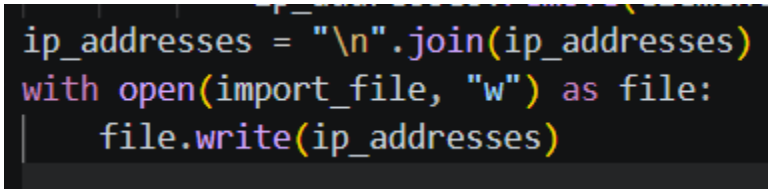
Next we iterate through the remove list, creating a variable called element for each IP address contained in the list

Remove IP addresses that are on the remove list

```
if element in ip_addresses:
    ip_addresses.remove(element)
```

As we iterate through the list, the conditional statement checks if the item in the remove_list is contained in the allow_list. If it is, this item is removed from the allow_list.

Update the file with the revised list of IP addresses



In order to update the file with the new list of IP addresses, we first have to convert the list back to a string using the join command. The individual IP addresses are separated by the \n newline character so that each address is listed on a new line. We then open the file again as a write file this time, and write the string back to the file.

Summary

For this project I created an algorithm that reads the allowed IP addresses and the IP addresses that need to be removed, and removes these addresses from the original list. This then returns the corrected list to the original file of addresses.